Academic curriculum vitae



Panagiotis Tsoutsanis

Professor of Computational Fluid Dynamics
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ORCID · Google Scholar · Institutional profile · UCNS3D

Research overview

My research focuses on the numerical methods that make complex fluid flows computable: accurate reconstruction on unstructured meshes, stable treatment of shocks and material interfaces, and efficient resolution of turbulence. I develop and analyse high-order finite-volume and discontinuous Galerkin methods, with particular attention to how numerical dissipation and dispersion affect the physics captured by a simulation.

This work connects algorithm development with applications in compressible, multiphase and high-speed flows. Through UCNS3D, I bring these methods into an open-source simulation framework and investigate how they can run efficiently on both CPUs and GPUs. The aim is to make accurate, robust CFD practical for complex geometries and large-scale calculations.

Areas of expertise

High-order methods on unstructured meshes. Finite-volume and discontinuous Galerkin formulations, accurate reconstruction and mixed-element meshes.

Shock capturing and adaptive methods. WENO and CWENO schemes, a posteriori MOOD limiting, and adaptive control of numerical dissipation and dispersion.

Turbulence and aerodynamics. Implicit large-eddy simulation, RANS modelling and the influence of numerical choices on aerodynamic predictions.

Multiphase and multicomponent flows. Diffuse-interface capturing, shock–interface interactions, cavitation and complex transient flows.

High-speed and hypersonic flows. Compressible-flow algorithms, shock–boundary-layer interactions and aerodynamic and thermal prediction.

High-performance scientific computing. Parallel algorithms, CPU and GPU execution, performance portability and open research software through UCNS3D.

Education and professional development

Professional Certificate in Quantum Computing Fundamentals
MIT xPRO
2024

Postgraduate Certificate in Academic Practice in Higher Education
Cranfield University
2017–2018

PhD in Computational Fluid Dynamics
Cranfield University
Thesis: Very high-order methods for 3D arbitrary unstructured grids
2006–2009

MEng in Aerospace Engineering
University of Liverpool
1998–2003

Selected distinctions

ARCHER2 Image and Video Competition — Winning Video
Shock wave interaction with cavitation bubble
2021
Award record

ARCHER Image and Video Competition — Winning Early Career Researcher Entry
Supersonic Kelvin–Helmholtz instability simulation
2017
Award record

Young Scientist Award, HONOM / ECCOMAS
High-order numerical methods for mixed-element unstructured meshes
2009

Research software

UCNS3D is an open-source framework for high-order computational fluid dynamics on unstructured meshes. I am its author and lead developer.

ucns3d.com · Source code

Teaching and supervision

My teaching connects numerical analysis, fluid physics and scientific software. It covers numerical methods for compressible flows and partial differential equations, high-performance computing, turbulence modelling, mesh generation and simulation practice.

I am a Fellow of the Higher Education Academy and hold a postgraduate certificate in Academic Practice in Higher Education.

Research students and alumni

Current research students

William Picarello
Multi-body hypersonic aerodynamics.
PhD · 2025–2029

Michal Uminski
Multidimensional optimal order detection techniques for arbitrary Lagrangian–Eulerian schemes for compressible flows.
PhD · 2024–2028

Cillian Wang
Multimodality sensing and control for nonlinear systems.
PhD · 2023–2027

Jacob Guthrie
Development of a novel short-range dispersal simulation capability.
PhD · 2023–2027

Yiren Tong
Development of high-order finite-volume methods for transonic aeronautical flows.
PhD · 2023–2026

Research alumni

Aaron Feria Alanis
Computational modelling of molecular dynamics in elastomeric seal leakage.
PhD · 2021–2025

Vadim Maltsev
High-order hybrid discontinuous Galerkin–finite-volume methods for multiphysics modelling.
PhD · 2024

Dean Yuan
High-order hybrid discontinuous Galerkin–finite-volume methods for aeronautical flows.
PhD · 2024

Ebenezer Adebayo Mayowa
Development of high-fidelity numerical methods for multicomponent and multiphase flows.
PhD · 2023

Paulo A.S.F. Silva
Development of a multiple reference frame technique for high-order CFD solvers.
PhD · 2021

Lun Ma
Renewable energy marine structures (REMS).
EngD · 2021

Diwakar Singh
Multi-fidelity, multi-objective optimisation of aircraft in formation flight.
PhD · 2019

Konstantinos Vasilopoulos
Airflow distribution and hazardous-release dispersion around a city using LES.
PhD · 2018

Zhongyuan Wang
Detached-eddy simulation for high-lift devices.
MRes · 2018

Nektarios Papantoniou
Assessment of tank projectile aerodynamics using CFD.
MRes · 2018

Nicolas Simmonds
Cooling-system optimisation for automotive vehicles.
PhD · 2017

Harshavardana Srinivasan
Adaptive mesh refinement using high-order WENO methods.
MPhil · 2017

Pablo Leandro Aguado
Large-eddy simulation of air–fuel mixing in a dry low-NOx GTU combustor.
MRes · 2014

Dimitris Nakos
High-resolution schemes for bluff-body aerodynamics.
MRes · 2013

Publications

Journal articles, preprints and reports, arranged by citation year. Publication status is retained where relevant.

2026

A hybrid finite-volume reconstruction framework for efficient high-order shock-capturing on unstructured meshes
Yiren Tong and Panagiotis Tsoutsanis
Computers & Fluids, 308, 106988
Journal article
DOI · Preprint

eMOOD: Entropy-Based Multidimensional Optimal Order Detection for High-Order Shock-Capturing Finite-Volume Schemes
Michał Umiński, Panagiotis Tsoutsanis, Ahmed A Sheikh Al-Shabab, and Alan S. Dawes
arXiv preprint
Preprint
DOI · Preprint

Towards Heterogeneous Exascale CFD with a Single Fortran Code Base: OpenMP Target Acceleration of the High-Order Unstructured Solver UCNS3D
Panagiotis Tsoutsanis, Michal Uminski, Christopher Day, and David Henty
arXiv preprint
Preprint
Preprint

2025

A Review of Diffuse Interface-Capturing Methods for Compressible Multiphase Flows
Ebenezer Mayowa Adebayo, Panagiotis Tsoutsanis, and Karl W. Jenkins
Fluids, 10(4), 93
Journal article
DOI

Arbitrary-order unstructured finite-volume methods for implicit large eddy simulation of turbulent flows with adaptive dissipation/dispersion adjustment (ADDA)
Panagiotis Tsoutsanis and Xesus Nogueira
Journal of Computational Physics, 523, 113653
Journal article
DOI

2024

A comprehensive CFD investigation of tip vortex trajectory in shrouded wind turbines using compressible RANS solver
Paulo A.S.F. Silva, Panagiotis Tsoutsanis, Jerson R.P. Vaz, and Marianela M. Macias
Energy, 294, 130929
Journal article
DOI

A Mixed-Elastohydrodynamic Lubrication Model of a Capped-T-Ring Seal with a Sectioned Multi-Material Film Thickness in Landing Gear Shock Absorber Applications
Aaron Feria Alanis, Ahmed A. Sheikh Al-Shabab, Antonis F. Antoniadis, Panagiotis Tsoutsanis, and Martin Skote
Fluids, 9(12), 271
Journal article
DOI

Application of Central-Weighted Essentially Non-Oscillatory Finite-Volume Interface-Capturing Schemes for Modeling Cavitation Induced by an Underwater Explosion
Ebenezer Mayowa Adebayo, Panagiotis Tsoutsanis, and Karl W. Jenkins
Fluids, 9(2), 33
Journal article
DOI

High-order hybrid DG-FV framework for compressible multi-fluid problems on unstructured meshes
Vadim Maltsev, Martin Skote, and Panagiotis Tsoutsanis
Journal of Computational Physics, 502, 112819
Journal article
DOI

Study of Orifice Design on Oleo-Pneumatic Shock Absorber
Paulo A. S. F. Silva, Ahmed A. Sheikh Al-Shabab, Panagiotis Tsoutsanis, and Martin Skote
Fluids, 9(5), 108
Journal article
DOI

Unsteady Multiphase Simulation of Oleo-Pneumatic Shock Absorber Flow
Ahmed A. Sheikh Al-Shabab, Bojan Grenko, Paulo A. S. F. Silva, Antonis F. Antoniadis, Panagiotis Tsoutsanis, and Martin Skote
Fluids, 9(3), 68
Journal article
DOI

2023

A relaxed a posteriori MOOD algorithm for multicomponent compressible flows using high-order finite-volume methods on unstructured meshes
Panagiotis Tsoutsanis, Machavolu Sai Santosh Pavan Kumar, and Pericles S. Farmakis
Applied Mathematics and Computation, 437, 127544
Journal article
DOI

A short note on a 3D spectral analysis for turbulent flows on unstructured meshes
Panagiotis Tsoutsanis, Xesus Nogueira, and Lin Fu
Journal of Computational Physics, 474, 111804
Journal article
DOI

Hybrid discontinuous Galerkin-finite volume techniques for compressible flows on unstructured meshes
Vadim Maltsev, Dean Yuan, Karl W. Jenkins, Martin Skote, and Panagiotis Tsoutsanis
Journal of Computational Physics, 473, 111755
Journal article
DOI

Numerical Investigation of the Inviscid Taylor-Green Vortex Using an Adaptive Filtering Method for a Modal Discontinuous Galerkin Method
Dean Yuan, Karl W. Jenkins, and Panagiotis Tsoutsanis
International Journal of Computational Fluid Dynamics, 37(6), 522–540
Journal article
DOI

2022

Blade-Resolved CFD Simulations of a Periodic Array of NREL 5 MW Rotors with and without Towers
Lun Ma, Pierre-Luc Delafin, Panagiotis Tsoutsanis, Antonis Antoniadis, and Takafumi Nishino
Wind, 2(1), 51–67
Journal article
DOI

High-order methods for diffuse-interface models in compressible multi-medium flows: A review
V. Maltsev, M. Skote, and P. Tsoutsanis
Physics of Fluids, 34(2), 021301
Journal article
DOI

Numerical investigation of full helicopter with and without the ground effect
Paulo A. S. F. Silva, Panagiotis Tsoutsanis, and Antonios F. Antoniadis
Aerospace Science and Technology, 122, 107401
Journal article
DOI

Numerical Investigation of Orifice Nearfield Flow Development in Oleo-Pneumatic Shock Absorbers
Ahmed A. Sheikh Al-Shabab, Bojan Grenko, Dimitrios Vitlaris, Panagiotis Tsoutsanis, Antonios F. Antoniadis, and Martin Skote
Fluids, 7(2), 54
Journal article
DOI

UCNS3D: An open-source high-order finite-volume unstructured CFD solver
Antonis F. Antoniadis, Dimitris Drikakis, Pericles S. Farmakis, Lin Fu, Ioannis Kokkinakis, Xesús Nogueira, Paulo A. S. F. Silva, Martin Skote, Vladimir Titarev, and Panagiotis Tsoutsanis
Computer Physics Communications, 279, 108453
Journal article
DOI

2021

A reduced-dissipation WENO scheme with automatic dissipation adjustment
Javier Fernández-Fidalgo, Luis Ramírez, Panagiotis Tsoutsanis, Ignasi Colominas, and Xesús Nogueira
Journal of Computational Physics, 425, 109749
Journal article
DOI

Arbitrary high order central non-oscillatory schemes on mixed-element unstructured meshes
Panagiotis Tsoutsanis and Michael Dumbser
Computers & Fluids, 225, 104961
Journal article
DOI

CWENO Finite-Volume Interface Capturing Schemes for Multicomponent Flows Using Unstructured Meshes
Panagiotis Tsoutsanis, Ebenezer Mayowa Adebayo, Adrian Carriba Merino, Agustin Perez Arjona, and Martin Skote
Journal of Scientific Computing, 89(3), 64
Journal article
DOI

Large eddy simulation of dispersion of hazardous materials released from a fire accident around a cubical building
Konstantinos Vasilopoulos, Ioannis Lekakis, Ioannis E. Sarris, and Panagiotis Tsoutsanis
Environmental Science and Pollution Research, 28(36), 50363–50377
Journal article
DOI

Simple multiple reference frame for high-order solution of hovering rotors with and without ground effect
Paulo A.S.F. Silva, Panagiotis Tsoutsanis, and Antonis F. Antoniadis
Aerospace Science and Technology, 111, 106518
Journal article
DOI

2020

Hovering rotor solutions by high-order methods on unstructured grids
Francesco Ricci, Paulo A.S.F. Silva, Panagiotis Tsoutsanis, and Antonis F. Antoniadis
Aerospace Science and Technology, 97, 105648
Journal article
DOI

WENO schemes on unstructured meshes using a relaxed a posteriori MOOD limiting approach
Pericles S. Farmakis, Panagiotis Tsoutsanis, and Xesús Nogueira
Computer Methods in Applied Mechanics and Engineering, 363, 112921
Journal article
DOI

2019

Assessment of air flow distribution and hazardous release dispersion around a single obstacle using Reynolds-averaged Navier-Stokes equations
Konstantinos Vasilopoulos, Ioannis E. Sarris, and Panagiotis Tsoutsanis
Heliyon, 5(4), e01482
Journal article
DOI

Optimisation of the Higher-Order Finite-Volume Unstructured Code Enhancement for Compressible Turbulent Flows
A. Shamakina and Panagiotis Tsoutsanis
Partnership for Advanced Computing in Europe (PRACE)
Report
Publication / event record

Stencil selection algorithms for WENO schemes on unstructured meshes
Panagiotis Tsoutsanis
Journal of Computational Physics: X, 4, 100037
Journal article
DOI

2018

A Multi-Fidelity Approach for Aerodynamic Performance Computations of Formation Flight
Diwakar Singh, Antonios F. Antoniadis, Panagiotis Tsoutsanis, Hyo-Sang Shin, Antonios Tsourdos, Samuel Mathekga, and Karl W. Jenkins
Aerospace, 5(2), 66
Journal article
DOI

Computational Assessment of the Hazardous Release Dispersion from a Diesel Pool Fire in a Complex Building's Area
Konstantinos Vasilopoulos, Michalis Mentzos, Ioannis E. Sarris, and Panagiotis Tsoutsanis
Computation, 6(4), 65
Journal article
DOI

Extended bounds limiter for high-order finite-volume schemes on unstructured meshes
Panagiotis Tsoutsanis
Journal of Computational Physics, 362, 69–94
Journal article
DOI

Improvement of the computational performance of a parallel unstructured WENO finite volume CFD code for Implicit Large Eddy Simulation
Panagiotis Tsoutsanis, Antonis F. Antoniadis, and Karl W. Jenkins
Computers & Fluids, 173, 157–170
Journal article
DOI

Low-Mach number treatment for Finite-Volume schemes on unstructured meshes
Nicholas Simmonds, Panagiotis Tsoutsanis, Antonis F. Antoniadis, Karl W. Jenkins, and Adrian Gaylard
Applied Mathematics and Computation, 336, 368–393
Journal article
DOI

2017

Assessment of high-order finite volume methods on unstructured meshes for RANS solutions of aeronautical configurations
Antonis F. Antoniadis, Panagiotis Tsoutsanis, and Dimitris Drikakis
Computers & Fluids, 146, 86–104
Journal article
DOI

Complete Body Aerodynamic Study of three Vehicles
Nicholas Simmonds, John Pitman, Panagiotis Tsoutsanis, Karl Jenkins, Adrian Gaylard, and Wilko Jansen
SAE International, 2017-01-1529
Report
DOI

KNL Performance Comparison: UCNS3D
Panagiotis Tsoutsanis and Antonios Antoniadis
ARCHER
Report
Publication / event record

2016

A high-order finite-volume method for atmospheric flows on unstructured grids
Panagiotis Tsoutsanis and Dimitris Drikakis
Journal of Coupled Systems and Multiscale Dynamics, 4(3), 170–186
Journal article
DOI

Full Vehicle Aero-Thermal Cooling Drag Sensitivity Analysis for Various Radiator Pressure Drops
Nicholas Simmonds, Panagiotis Tsoutsanis, Dimitris Drikakis, Adrian Gaylard, and Wilko Jansen
SAE International, 2016-01-1578
Report
DOI

Optimising UCNS3D, a High-Order finite-Volume WENO Scheme Code for arbitrary unstructured Meshes
Thomas Ponweiser and Panagiotis Tsoutsanis
Partnership for Advanced Computing in Europe (PRACE)
Report
Publication / event record

2015

Comparison of structured- and unstructured-grid, compressible and incompressible methods using the vortex pairing problem
Panagiotis Tsoutsanis, Ioannis W. Kokkinakis, László Könözsy, Dimitris Drikakis, Robin J.R. Williams, and David L. Youngs
Computer Methods in Applied Mechanics and Engineering, 293, 207–231
Journal article
DOI

2014

WENO schemes on arbitrary unstructured meshes for laminar, transitional and turbulent flows
Panagiotis Tsoutsanis, Antonios Foivos Antoniadis, and Dimitris Drikakis
Journal of Computational Physics, 256, 254–276
Journal article
DOI

2011

WENO schemes on arbitrary mixed-element unstructured meshes in three space dimensions
P. Tsoutsanis, V.A. Titarev, and D. Drikakis
Journal of Computational Physics, 230(4), 1585–1601
Journal article
DOI

2010

WENO Schemes for Mixed-Element Unstructured Meshes
V. A. Titarev, P. Tsoutsanis, and D. Drikakis
Communications in Computational Physics, 8(3), 585–609
Journal article
DOI

Conference contributions

Conference papers and presentations, arranged by citation year. An event year is shown separately when it differs.

2025

A Modular Multifidelity Approach for Multiphysics Oleo-Pneumatic Shock Absorber Simulations
Ahmed A. Sheikh Al-Shabab, Paulo A. S. F. Silva, Bojan Grenko, Panagiotis Tsoutsanis, and Martin Skote
Proceedings of the Cambridge Unsteady Flow Symposium 2024, 137–151
Conference paper · Event year: 2024
DOI

2024

A Multifidelity Framework For Oleo-Pneumatic Shock Absorber Simulations
A.A.S. Al-Shabab, P. A.S.F. Silva, P. Tsoutsanis, and M. Skote
Cambridge Unsteady Flow Symposium
Conference presentation

Adaptive Viscosity Methods for Turbulent Compressible Flow Computations
Xesús Nogueira, Javier Fernández Fidalgo, Luis Ramírez, and Panagiotis Tsoutsanis
ECCOMAS 2024 Congress
Conference presentation
Publication / event record

Adaptively Tuned High-Order Methods for iLES
Panagiotis Tsoutsanis, Lin Fu, and Xesus Nogueira
ECCOMAS Congress 2024: 9th European Congress on Computational Methods in Applied Sciences and Engineering
Conference paper
DOI

ADDAptive Numerical Framework for iLES of Compressible Flows
P. Tsoutsanis
HONOM 2024 Conference
Conference presentation

Automated Flow Physics Identification and Classification in Multiphysics Multifidelity Simulations
Ahmed A. Sheikh Al-Shabab, Paulo A. S. F. Silva, Panagiotis Tsoutsanis, and Martin Skote
ECCOMAS Congress 2024: 9th European Congress on Computational Methods in Applied Sciences and Engineering
Conference paper
DOI

Comparative Study of Orifice Shapes on Shock Absorber: Impact on Energy Dissipation and Cavitation
Paulo A. S. F. Silva, Ahmed A. Sheikh Al-Shabab, Panagiotis Tsoutsanis, and Martin Skote
ECCOMAS Congress 2024: 9th European Congress on Computational Methods in Applied Sciences and Engineering
Conference paper
DOI

Computational Analysis of a High-Compression Hydraulic Landing Gear System for Enhanced Aircraft Safety
Y. Wei and P. Tsoutsanis
19th OpenFoam Conference
Conference presentation

Dynamic swirl distortion characteristics in S-shaped diffusers using UCNS3D and time-resolved, stereo PIV methods
Tommaso Piovesan, Matteo Migliorini, Pavlos K. Zachos, and Panagiotis Tsoutsanis
AIAA SCITECH 2024 Forum
Conference paper
DOI

2023

A hybrid DG/FV framework for gas-gas and gas-liquid compressible flows
Vadim Maltsev, Martin Skote, and Panagiotis Tsoutsanis
CFC 2023 Conference
Conference presentation
Publication / event record

A Mixed-Elastohydrodynamic Lubrication Model of a Capped-T-ring Seal Based on a Modified Reynolds Equation
Aaron Feria Alanis, Martin Skote, Panagiotis Tsoutsanis, and Antonis Foivos Antoniadis
CFC 2023 Conference
Conference presentation
Publication / event record

Adaptive Filtering for Hybrid DG-FV methods for Turbulent flows on Unstructured Meshes
Dean Yuan, Panagiotis Tsoutsanis, and Xesus Nogueira
CFC 2023 Conference
Conference presentation
Publication / event record

Automatic Dissipation/Dispersion Control for High-Order Finite-Volume Schemes using Unstructured Meshes for Turbulent Flows
Panagiotis Tsoutsanis, Dean Yuan, and Xesus Nogueira
CFC 2023 Conference
Conference presentation
Publication / event record

Multi-Physics and Multi-Fidelity Considerations in Oleo-Pneumatic Shock Absorber Orifice Flow
Ahmed A. Sheikh Al-Shabab, Bojan Grenko, Antonis Antoniadis, Panagiotis Tsoutsanis, and Martin Skote
CFC 2023 Conference
Conference presentation
Publication / event record

Tip-vortex interaction on Shrouded Wind-Turbines
Paulo Silva, Panagiotis Tsoutsanis, and Karl Jenkins
CFC 2023 Conference
Conference presentation
Publication / event record

2022

A posteriori MOOD limiting approach for multicomponent flows on unstructured meshes
P. Tsoutsanis and P. S. Farmakis
ECCOMAS Congress 2022: 8th European Congress on Computational Methods in Applied Sciences and Engineering
Conference paper
DOI

An adaptive-dissipation WENO scheme for the numerical simulation of compressible turbulent flows
Javier Fernandez-Fidalgo, Luis Ramirez, P. Tsoutsanis, Ignasi Colominas, and Xesus Nogueira
HONOM 2022 Conference
Conference presentation

Application of interface capturing schemes on multiphase/multicomponent compressible flow of underwater explosion
E. M. Adebayo, P. Tsoutsanis, and K. W. Jenkins
ECCOMAS Congress 2022: 8th European Congress on Computational Methods in Applied Sciences and Engineering
Conference paper
DOI

Hybrid Finite-Volume/Discontinuous Galerkin framework for the solution of Multiphysics problems using unstructured meshes
V. Maltsev, P. Tsoutsanis, and M. Skote
ECCOMAS Congress 2022: 8th European Congress on Computational Methods in Applied Sciences and Engineering
Conference paper
DOI

Hybrid High-Order Finite Volume/Discontinuous Galerkin Methods for Turbulent Flows
Dean Yuan, Panagiotis Tsoutsanis, and Karl W. Jenkins
ECCOMAS Congress 2022: 8th European Congress on Computational Methods in Applied Sciences and Engineering
Conference paper
DOI

Numerical Investigation of Oleo-Pneumatic Shock Absorber: A Multifidelity Approach
Ahmed A. Sheikh Al-Shabab, Bojan Grenko, Dimitrios Vitlaris, Panagiotis Tsoutsanis, Antonis F. Antoniadis, and Martin Skote
ECCOMAS Congress 2022: 8th European Congress on Computational Methods in Applied Sciences and Engineering
Conference paper
DOI

UCNS3D:A High-Order Finite-Volume CFD solver for Compressible flows
P. Tsoutsanis
SHARK FV 2022 Conference
Conference presentation

Unstructured high-order solutions of hovering rotors with and without ground effect
Paulo A. S. F. Silva, Panagiotis Tsoutsanis, Antonis F. Antoniadis, and Karl W. Jenkins
ECCOMAS Congress 2022: 8th European Congress on Computational Methods in Applied Sciences and Engineering
Conference paper
DOI

2021

Multiphysics Unsteady Simulation of an Oleo Pneumatic Shock Absorber
A. A. S. Al-Shabab, Z. Liu, P. Tsoutsanis, A. F. Antoniadis, and M. Skote
14th World Congress on Computational Mechanics and ECCOMAS Congress 2020
Conference presentation

Numerical Investigation of Oleo-Pneumatic Shock Absorber: Setup and Validation
Ahmed A. Sheikh Al-Shabab, Dimitrios Vitlaris, Zhonglu Lin, Bojan Grenko, Panagiotis Tsoutsanis, Antonis Antoniadis, and Martin Skote
14th World Congress on Computational Mechanics and ECCOMAS Congress 2020
Conference paper
DOI

Numerical Simulation of Oleo Pneumatic Shock Absorber: Validation and Case Setup
Z. Liu, A. A. S. Al-Shabab, P. Tsoutsanis, A. F. Antoniadis, and M. Skote
14th World Congress on Computational Mechanics and ECCOMAS Congress 2020
Conference presentation

Withdrawn: Effects of Heat-Conductive Obstacles on Conjugate Heat Transfer of Backward-Facing Step Flow
Zhonglu Lin, Ahmed A. Al-Shabab, Dimitrios Vitlaris, Panagiotis Tsoutsanis, Antonis Antoniadis, and Martin Skote
AIAA Scitech 2021 Forum
Conference paper · WITHDRAWN
DOI

2019

Computational Aerodynamics Solutions of Hovering Rotors by high-order schemes on unstructured grids
P. A.S.F. Silva, F. Ricci, P. Tsoutsanis, K. W. Jenkins, and A. F. Antoniadis
UK FLUIDS 2019
Conference presentation

Diesel Pool Fire Incident Inside an Urban Street Canyon
Konstantinos Vasilopoulos, Ioannis E. Sarris, Ioannis Lekakis, and Panagiotis Tsoutsanis
Proceedings of the 1st International Conference on Numerical Modelling in Engineering, 339–350
Conference paper · Event year: 2018
DOI

2018

High-Order Accurate Reconstruction Scheme for the Rayleigh-Taylor and Richtmyer-Meshkov Instabilities Using Arbitrary Unstructured Meshes
Pericles S. Farmakis and Panagiotis Tsoutsanis
11th Panhellenic Conference on Fluid Mechanics
Conference paper
Publication / event record

2017

Improvement of the Computational Performance of a Parallel Unstructured WENO Finite Volume CFD code on Knights Landing Processors
A. F. Antoniadis, K.W. Jenkins, and P. Tsoutsanis
ParCFD conference
Conference presentation

Improving the Curing Cycle Time through the Numerical Modeling of air Flow in Industrial Continuous Convection Ovens
Yuan Yi, Konstantinos Salonitis, Panagiotis Tsoutsanis, Lampros Litos, and John Patsavelas
Procedia CIRP, 63, 499–504
Conference paper
DOI

2016

Adaptive mesh refinement techniques for high-order finite-volume WENO scheme
Harshavardhana Srinivasan and Panagiotis Tsoutsanis
Proceedings of the VII European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS Congress 2016), 2883–2899
Conference paper
DOI

Addressing the Challenges of Implementation of High-Order Finite-Volume Schemes for Atmospheric Dynamics on Unstructured Meshes
Panagiotis Tsoutsanis and Dimitris Drikakis
Proceedings of the VII European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS Congress 2016), 684–708
Conference paper
DOI

Implementation of a low-Mach number modification for High-Order Finite-Volume Schemes for Arbitrary Hybrid Unstructured Meshes
Nicholas Simmonds, Panagiotis Tsoutsanis, and Adrian Gaylard
Proceedings of the VII European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS Congress 2016), 2900–2915
Conference paper
DOI

2015

A Multi-Fidelity, Multi-Disciplinary Analysis and Optimization Framework for the Design of Morphing UAV Wing
Andrea Ciarella, Christos Tsotskas, Marco Hahn, Noud Werter, Roeland De Breuker, Chris Beaverstock, Michael I. Friswell, Yosheph Yang, Serkan Özgen, Antonios Antoniadis, Dimitris Drikakis, and Panagiotis Tsoutsanis
16th AIAA/ISSMO Multidisciplinary Analysis and Optimization Conference
Conference paper
DOI

Azure: An Advanced CFD Software Suite Based on High-Resolution and High-Order Methods
Dimitris Drikakis, Antonios F. Antoniadis, Panagiotis Tsoutsanis, Ioannis Kokkinakis, and Zeeshan A. Rana
53rd AIAA Aerospace Sciences Meeting
Conference paper
DOI

High-order methods for hypersonic shock wave turbulent boundary layer interaction
Ioannis W. Kokkinakis, Antonios F. Antoniadis, Dimitris Drikakis, Panagiotis Tsoutsanis, and Zeeshan A. Rana
20th AIAA International Space Planes and Hypersonic Systems and Technologies Conference, 262–280
Conference paper
DOI

Numerical Accuracy in RANS Computations of High-Lift Multi-element Airfoil and Aircraft Configurations
Antonis F. Antoniadis, Panagiotis Tsoutsanis, and Dimitris Drikakis
53rd AIAA Aerospace Sciences Meeting
Conference paper
DOI

2014

High-order RANS Solutions for Full Aircraft and High-Lift Devices
A. F. Antoniadis, P. Tsoutsanis, and D. Drikakis
Proceeding of the Royal Aeronautical Society Applied Aerodynamics Conference
Conference presentation

2012

High-order Schemes on Mixed-element Unstructured Grids for Aerodynamic Flows
Antonios Antoniadis, Panagiotis Tsoutsanis, and Dimitrios Drikakis
42nd AIAA Fluid Dynamics Conference and Exhibit
Conference paper
DOI

Large eddy simulation of turbulent flows on 3D hybrid unstructured meshes
P. Tsoutsanis, D. Drikakis, and J. Lelieveld
ECCOMAS 2012 Congress
Conference presentation

Modelling atmospheric flows on 3D hybrid unstructured meshes using high-order methods
P. Tsoutsanis, D. Drikakis, and J. Lelieveld
3rd International EULAG Workshop
Conference presentation

2009

WENO Methods on 3D Unstructured Hexahedral Meshes
P. Tsoutsanis, V. Titarev, and D. Drikakis
HONOM2009 ECCOMAS Thematic Conference
Conference presentation

2008

Very High-Order Godunov-Type Methods on Unstructured Hexahedral Meshes in Three Space Dimensions
P. Tsoutsanis, V. Titarev, and D. Drikakis
ECCOMAS 2008 Congress
Conference presentation

Invited talks and seminars

ADDAptive Numerical Framework for iLES of Compressible Flows
HONOM 2024 · Chania, Greece
September 2024 · Keynote

High-Order Non-Linear Methods on Unstructured Meshes for Compressible Flows
Hong Kong University of Science and Technology · Hong Kong
November 2023 · Invited seminar

Compact Methods for Compressible Flows
University of Waterloo · Waterloo, Canada
November 2023 · Invited seminar

High-Order Methods for Compressible Flows
Department of Physics, Cavendish Laboratory, University of Cambridge · Cambridge, UK
March 2023 · Invited seminar

CFD Research Capabilities at ANM Group
AWE · UK
October 2022 · Invited seminar

Advancement of Adaptive Dissipation for Turbulent Flows
University of A Coruña · A Coruña, Spain
October 2022 · Invited seminar

UCNS3D: An Open-Source High-Order Finite-Volume Unstructured CFD Solver
Computer Physics Communications Seminar Series
October 2022 · Invited seminar
Recording

UCNS3D: An Open-Source High-Order Framework: Philosophy and Future
CRUNCH Seminar Series, Brown University · Providence, USA
September 2022 · Invited seminar

UCNS3D: An Overview of the Methods and Capabilities
SHARK-FV workshop · Porto, Portugal
June 2022 · Workshop talk
Event

Turbulence at the Exascale
Excalibur podcast
June 2021 · Podcast
Recording

UCNS3D: A High-Order Compressible Open Source CFD: Past, Present and Future
International Society for CFD
April 2021 · Invited seminar
Recording

Stencil Selection Algorithm for Unstructured Meshes
Barcelona Supercomputing Centre · Barcelona, Spain
March 2020 · Invited seminar

Compute for Cosmos
Department of Physics, Cavendish Laboratory, University of Cambridge · Cambridge, UK
December 2019 · Invited seminar

Optimisation of High-Order Finite Volume Methods for Compressible Flows
Perm University · Perm, Russia
November 2013 · Invited seminar

Addressing the Challenges of Implementation of High-Order Finite-Volume Schemes for Atmospheric Dynamics on Unstructured Meshes
Loughborough University · Loughborough, UK
December 2012 · Invited seminar

MSc research topics

Topics from individual and group MSc research projects, organised by academic year.

2023-24

CFD modelling of H_2 deflagrations in enclosures
MSc in CFD

Decomposition of an organic pollutant by photocatalysis within a Taylor–Couette flow reactor
MSc in CFD

CFD modelling of liquid springs for landing gear
MSc in CFD

Advancing hypersonic modelling with high-order methods using UCNS3D
MSc in CFD

Effects of bleed-plenum geometry on boundary-layer formation
MSc in CFD

High-fidelity CFD modelling of stealth intakes
MSc in CFD

Finite-element analysis: correlation between simulation and experimental testing
MSc in ACE

Creation and calibration of different types of soil
MSc in ACE

Development and application of bleed control for supersonic intakes using UCNS3D
MSc in ACE

Mesh generation for numerical simulation in aerodynamics
MSc in ACE

CFD modelling of liquid springs
MSc in ACE

Sloshing effects in a cryogenic hydrogen-storage tank for commercial aircraft
MSc in CFD

Air–hydrogen mixing modelling in a dry low-NOx GTU combustor
MSc in CFD

2022-23

CFD simulation of a closing valve through space- and time-dependent porosity convection: a preliminary study for fluid–structure interaction in two-phase flows
MSc in CFD

Using CFD to assess the hydrodynamic impact of the implementation and strategic placement of flow injection over a flapped rudder
MSc in CFD

RANS modelling of HiFIRE hypersonic geometry
MSc in ACE

Numerical simulation of cavitation on rotors
MSc in CFD

Impact of numerical schemes and grid design on oil–gas interface monitoring in landing-gear shock absorbers
MSc in CFD

Cavitation modelling for landing-gear shock absorbers
MSc in CFD

Assessment of high-order Godunov-type numerical methods in the strong-shock regime
MSc in CFD

CFD analysis of a supersonic civil airliner
MSc in ACE

2021-22

AMR for CFD modelling of sonic booms for supersonic civil airliners
MSc in CFD

Development of new robust Riemann solvers for high-speed flows for nuclear fusion
MSc in CFD

Optimisation of existing Riemann solvers for high-speed flows found in nuclear fusion
MSc in CFD

Target-design modelling for nuclear fusion using high-order finite-volume WENO methods
MSc in CFD

Target-design modelling for nuclear fusion using the MOOD framework with UCNS3D
MSc in CFD

Sonic-boom prediction for civil supersonic cruisers using RANS simulations
MSc in CFD

Fluid–structure interaction of rapid aircraft decompression in a civil airliner
MSc in ACE

CFD modelling of the SpaceX heavy launch vehicle with high-order Euler methods using UCNS3D
MSc in CFD

CFD modelling of the SpaceX heavy launch vehicle with RANS models
MSc in CFD

2020-21

Cooling optimisation for battery electric vehicles through duct-shape optimisation
MSc in CFD

Adjoint optimisation of constraining flows pertinent to motorsport racing
MSc in CFD

Numerical calculation and evaluation of extrapolation coefficients used in speed-powering predictions based on model tests
MSc in CFD

Cooling optimisation for battery electric vehicles through duct-size optimisation
MSc in CFD

CFD modelling of COVID spread in university lecture theatres and analysis of ventilation impact
MSc in CFD

2019-20

In situ visualisation for massively parallel simulations using UCNS3D
MSc in CFD

High-order discretisation optimisation for the ADIGMA flow solver
MSc in CFD

CFD modelling of rapid aircraft-cabin decompression using inviscid Euler equations
MSc in CFD

Ducting optimisation for battery electric vehicles using the SAE body model
MSc in CFD

Development of multiphysics models for liquid–gas compressible flows
MSc in CFD

Development of multiphysics models for gases using a compressible-flow solver for landing-gear shock absorbers
MSc in CFD

CFD for fish-farm modelling
MSc in CFD

Development of high-order discretisation for a Python traffic-flow modelling tool using high-resolution numerical methods
MSc in CFD

CFD investigation of the directional stability of high-speed craft
MSc in CFD

Rapid aircraft-decompression modelling
MSc in ACE

Influence of F1 wheel design under the 2021 FIA rules
MSc in CFD

F1 race-car aerodynamics: front-wing design under the 2021 FIA rules
MSc in CFD

2018-19

CFD simulation of rapid and explosive aircraft decompression
MSc in CFD

Development of a Python traffic-flow modelling tool using high-resolution numerical methods
MSc in CFD

2017-18

CFD simulation of the first stage of a deflector-jet servo-valve at nominal and reduced temperatures
MSc in CFD

High-order a posteriori limited reconstruction for Rayleigh–Taylor and Richtmyer–Meshkov flows
MSc in CFD

A single rotating reference frame in a high-resolution CFD solver
MSc in CFD

Application of hybrid LES–RANS methods to a realistic car model in realistic on-road conditions
MSc in CFD

Influence of wheel modelling for the DrivAer model under realistic motorway conditions
MSc in CFD

Numerical simulation of the estate DrivAer model under realistic motorway conditions
MSc in CFD

2016-17

A simple rotated-hybrid HLL–HLLC Riemann solver for curing carbuncle and shock instability on high-order unstructured meshes for supersonic and hypersonic flows
MSc in CFD

Numerical investigation of flow past a generic notchback geometry under side-wind conditions
MSc in CFD

Computational investigation of severe crosswind influence on the fastback DrivAer car model using DDES
MSc in CFD

CFD simulation of the interaction between a ship airwake and a helicopter
MSc in CFD

Aerodynamic effects of accreted ice shapes on a NACA 0012 aerofoil using CFD
MSc in CFD

Aerodynamic investigation of the DrivAer car model using DDES
MSc in CFD

2015-16

The impact of unsteadiness on uncertainty in automotive aerodynamic simulation using OpenFOAM
MSc in CFD

CFD analysis of mass-burning-rate and ventilation coupling in confined and ventilated enclosures
MSc in CFD

Analysis and comparison of dynamic parameters of an aerofoil undergoing plunging, pitching and flapping motion
MSc in CFD

Assessment of AMR techniques using unstructured meshes for RANS simulations
MSc in CFD

Assessment of scale-resolving turbulence-modelling approaches with implicit subgrid-scale models for CFD simulations of flow over bluff bodies
MSc in CFD

CFD simulation of interaction between a ship airwake and an aircraft
MSc in CFD

Development and optimisation of a medical device for decontamination and sterilisation of indoor spaces
MSc in CFD

Improving accuracy and computation time for hydrofoil simulations for the America's Cup at NUMECA International
MSc in CFD

2014-15

Evaluation of the Spalart correction term for different turbulence models
MSc in CFD

Impact of curvature correction on wingtip-vortex flow
MSc in CFD

Implementation of the Chien low-Re k–ε turbulence model in UCNS3D
MSc in CFD

Implementation of the Wilcox k–ω turbulence model in UCNS3D
MSc in CFD

Slope limiters for higher-order methods on unstructured grids
MSc in CFD

Simulation of two-dimensional unsteady flow past a heated cylinder in a laminar vortex-shedding regime, Re = 80–140
MSc in CFD

Aerodynamic performance prediction using a sweeping method: improving the efficiency of RANS near stall
MSc in CFD

2013-14

Effects of numerical discretisation in vortical flows
MSc in CFD

Performance assessment of a higher-order scheme turbulence model for automotive flow
MSc in CFD

Drag prediction of a full aircraft configuration using higher-order methods
MSc in CFD

Modern parallelisation strategies for CFD applications
MSc in CFD

Low-Reynolds-number separation around aerofoils
MSc in CFD

Design-of-experiments-based optimisation for hydrofoils
MSc in CFD

Turbulent mixing in a gas-turbine combustor using high-order discretisation methods
MSc in CFD

Impact of numerics on turbulent mixing in a dry low-NOx gas-turbine-unit combustor
MSc in CFD

Assessment of high-order CFD methods against experimental flow measurements for automotive applications
MSc in CFD

Numerical study of shock-wave/boundary-layer interaction at Mach 5
MSc in CFD

Detailed description of the vortex topology inside simplified wheelhouses
MSc in CFD

2012-13

Investigation of high-order schemes on unstructured grids for shock-wave/turbulent-boundary-layer interaction flows
MSc in CFD

Numerical simulation of the HyShot-2 scramjet configuration using high-order schemes
MSc in CFD

Drag evaluation on a Windsor body with wheels within wheel housings
MSc in CFD

Interaction of a Formula 1 car front wing and an exposed wheel
MSc in CFD

Aerodynamic force modification through base roughness
MSc in CFD